

What Kind of Layoff Does the Market Punish?

Materiality, Motive, and the Limits of Administrative Data in a WARN-Notice Event Study

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Abstract

Firms lay off workers for very different reasons—to cut costs in a downturn, or to reallocate away from a shrinking line of business—and it is not obvious that equity investors treat the two alike. We build a multi-state panel of mass-layoff (WARN Act) notices for 2022–2025 from the California, Texas, and Oregon WARN databases (6,560 notices), fuzzy-match the filers to listed firms, and estimate market-model cumulative abnormal returns (CARs) around 615 firm-level layoff events at 328 public companies.

On average the market gives a *modest but statistically significant negative* verdict:

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CAR[0, +5] = -1.6% ($t = -2.41$), concentrated in the days after the notice, with a clean near-zero pre-event placebo (-0.18%, $t = -0.32$). But the average hides sharp heterogeneity, and it is not about *how many* people are let go. Absolute headcount is irrelevant ($t = 0.46$); what matters is the layoff’s size *relative to the firm* and, equivalently, how small the firm is. Small firms lose 3.9% ($t = -2.86$) while large firms are essentially unaffected (+0.4%, $t = 1.13$); a one-standard-deviation larger relative layoff moves CAR by -3.3 percentage points ($t = -2.77$), and this punishment falls almost entirely on small firms (size×relative-layoff interaction = -0.065, $t = -2.86$). Four journal-track additions sharpen the reading. Randomization inference (2,000 permutations) places the materiality slope far in the tail (design-based $p = 0.001$), so the effect does not rest on clustered-standard-error asymptotics. The average reaction survives three alternative abnormal-return benchmarks and is, if anything, *stronger* against market- and S&P-adjusted returns (-2.0%, $t = -3.0$). Crucially, our crude machine-readable proxy for *motive*—whether the filing is a plant *closure* (exit/reallocation) or an ongoing-operation *layoff* (cost-cutting)—shows *no* difference in CARs ($t = -0.07$; randomization $p = 0.92$), and a minimum-detectable-effect analysis shows this null is *not* a tight bound: the machine proxy could only have detected a motive effect of 5.1 percentage points at 80% power, far above plausible layoff-CAR magnitudes. The machine motive flag is thus both a point-null *and* too imprecise to rule out a real motive effect. Distinguishing cost-cutting from strategic reallocation therefore requires reading the announcements and hand-coding motive, which is the central task this paper sets up.

JEL: G14, J63, M51, G34. **Keywords:** mass layoffs, WARN Act, event study, abnormal returns, downsizing, restructuring, randomization inference, statistical power.

1 Introduction

When a firm announces that it is cutting jobs, the stock can go up or down, and which way it goes has long been read as a window into *why* the firm is cutting. A layoff undertaken to trim fat and defend margins may be cheered as discipline; a layoff forced by collapsing demand is a bad-news signal about the franchise; and a layoff that reallocates resources out of a dying business toward a growing one is something else again (Cascio, 1993; Love and Nohria, 2005). The empirical record on the average reaction is famously mixed—early studies found small negative reactions (Worrell et al., 1991; Palmon et al., 1997), later work found the reaction attenuating or turning positive over time (Farber and Hallock, 2009), and the sign appears to depend on the layoff’s stated *cause* (Elayan et al., 1998; Nixon et al., 2004; Chen et al., 2001). This paper asks a sharp version of the question using a data source that has been under-used in finance: the mandatory mass-layoff notices filed under the Worker Adjustment and Retraining Notification (WARN) Act (Addison and Blackburn, 1994).

Our question is: *across mass-layoff events at public firms, what kind of layoff does the market punish—and can the announced form of the layoff tell us the firm’s motive?* We answer it with a cross-sectional event study. We assemble every WARN notice filed in California, Texas, and Oregon from 2022 through 2025 (6,560 notices covering 3,450 distinct filers), fuzzy-match the filer names to CRSP/Compustat public firms, cluster each firm’s site-level notices into firm-level layoff *events*, and estimate market-model CARs against the CRSP value-weighted index. The matched sample with valid returns is 615 firm-events at 328 listed companies over 2022–2024.¹

Our findings are threefold, and each is then stress-tested with a design-based method. *First*, the average market reaction to a WARN notice is negative and significant, but small. Over the $[0, +5]$ window the mean CAR is -1.6% ($t = -2.41$; Table 2); the $[0, +2]$ window

¹The CRSP daily market index in our WRDS extract runs through 2024-12-31, so events with a full post-event window are those through December 2024; the WARN panel itself extends into 2025.

is -1.3% ($t = -2.22$); the reaction sits after the notice date, and the pre-event $[-10, -2]$ placebo is an insignificant -0.18% . Notably only 49% of CARs are negative: the mean is driven by a negative *tail*, not a uniform markdown. The market does not reflexively punish layoffs.

Second, what determines who lands in that negative tail is *firm size / relative layoff size*, not the raw number of jobs. The absolute headcount laid off carries no information for returns (Table 4 column 2, $t = 0.46$). The layoff’s size *relative to the firm’s workforce* does: a one-standard-deviation increase in the relative-layoff ratio is associated with a -3.3 -percentage-point CAR ($t = -2.77$, column 1). Equivalently, larger firms are spared: a one-SD increase in $\ln(\text{assets})$ moves CAR by $+3.1\text{pp}$ ($t = 3.34$, column 3). The two are two sides of one coin—a WARN filing is a locally large but firm-level tiny event for a mega-cap, and a material event for a small company. Splitting the sample makes this vivid (Table 3): small firms lose 3.9% ($t = -2.86$) and large firms gain a statistically-zero 0.4% ; top-half relative layoffs lose 3.5% ($t = -2.70$) and bottom-half relative layoffs are flat. The punishment is concentrated where the layoff is big relative to the firm: the $\text{size} \times \text{relative-layoff}$ interaction is -0.065 ($t = -2.86$; Table 5).

Third, and most important for the paper’s title question, our machine-readable proxy for *motive* does *not* separate winners from losers. State WARN filings flag whether a layoff is a permanent plant *closure*—a natural proxy for exit/reallocation—or an ongoing-operation *layoff*—a natural proxy for cost-cutting. Closures and ongoing-operation layoffs earn statistically identical CARs (-1.5% vs. -1.6% ; the closure dummy is -0.001 with $t = -0.07$ in the full specification, Table 4 column 5). The form of the layoff, as coded from the filing’s category field, is uninformative about motive in a way that moves prices.

We then push each result past the point where most short event studies stop, using four design-based additions that are the methodological contribution of this version of the paper. (i) *Randomization inference* (Section 8, Table 7, Figure 3): permuting the materiality regressor 2,000 times places the actual slope far outside the permutation distribution

(design-based $p = 0.001$), while the closure/motive coefficient sits dead-center ($p = 0.92$)—the materiality effect is real and the machine-motive null is robust to design-based inference, not an artifact of clustered-SE asymptotics with a moderate number of firms. (ii) *Alternative abnormal-return benchmarks* (Section 9, Table 8): re-deriving CARs from the raw daily returns under market-adjusted, S&P-500-adjusted, and mean-adjusted models leaves the average negative reaction and the materiality slope intact—and the market- and S&P-adjusted reactions are actually *larger* (-2.0% , $t \approx -3.0$). (iii) *Expanded heterogeneity on observable moderators* (Section 7, Table 6): the materiality effect does not concentrate in any observable subgroup (valuation, asset turnover, sample year, goods-producing), reinforcing that it is a broad materiality phenomenon rather than a proxy for something firm-specific. (iv) *Power and minimum detectable effects* (Section 10, Table 9): the materiality estimate (-2.5pp size-controlled, -3.3pp unconditional) is comparable to its $\approx 3.9\text{pp}$ 80%-power minimum detectable effect—the design is powered to detect roughly the effect it finds—whereas the machine motive proxy could only have detected a motive effect of 5.1 percentage points at 80% power—larger than the entire average reaction and than any plausible motive effect. So the closure flag is not merely *zero*; it is *too coarse and too noisy* to rule out a real motive effect. This is exactly the wedge that motivates hand-coding: the difference between “we are closing this plant because the product is dying” (reallocation) and “we are closing this plant to cut costs” (retrenchment) is not in the WARN category field—it is in the announcement text and the surrounding disclosure, which a human must read. Resolving true motive, and re-running the cross-section on it, is the pivotal contribution we leave to the student researcher (STUDENT_TASKS.md).

We make four contributions. First, relative to the classic layoff event-study literature (Worrell et al., 1991; Palmon et al., 1997; Chen et al., 2001; Farber and Hallock, 2009), we bring a new and cleaner event source—mandatory WARN filings with precise dates—and show that in a modern (2022–2024) sample the average reaction is small and negative but the action is entirely in the *relative* size of the cut. Second, relative to the strategy

literature on downsizing motive (Cascio, 1993; Nixon et al., 2004; Love and Nohria, 2005; Guthrie and Datta, 2008), we show—and quantify with a power calculation—that the coarse closure-vs-layoff distinction available in administrative data is *not* a usable proxy for the cost-cutting-vs-reallocation motive that theory says should matter. Third, relative to work using WARN data (Addison and Blackburn, 1994), we provide, to our knowledge, one of the first matched WARN-to-CRSP event studies. Fourth, we treat both the headline effect and the null as quantitative objects, reporting design-based (randomization) p -values, alternative-benchmark robustness, and minimum detectable effects, so that readers can see what the design can and cannot rule out (Young, 2019; Abadie, 2020). Section 2 places the paper in the literature; Section 3 develops hypotheses; Section 4 describes the data and matching; Section 5 the research design and inference; Section 6 the main results; Section 7 heterogeneity; Section 8 randomization inference; Section 9 alternative benchmarks; Section 10 power and the informativeness of the motive null; Section 11 robustness; and Section 12 concludes.

2 Related Literature

This paper connects three strands. The first is the *layoff event-study* literature. Early studies of the stock-price reaction to layoff and plant-closing announcements generally found small negative average returns (Worrell et al., 1991; Palmon et al., 1997), with the reaction depending on the stated reason for the cut: layoffs framed as demand-driven or financially-distressed drew more negative reactions than layoffs framed as efficiency-improving or strategic reorganizations (Elayan et al., 1998; Chen et al., 2001; Nixon et al., 2004). Farber and Hallock (2009) document that the average reaction attenuated and at times turned positive over the 1970–1999 period, consistent with layoffs being increasingly read as proactive restructuring rather than distress. Our contribution to this strand is a new event source—mandatory WARN filings, which carry precise, legally-required dates and a site-level headcount—and a modern (2022–2024) sample in which we show that the average masks a materiality gradi-

ent: the reaction is entirely about the size of the cut *relative to the firm*, not the number of workers or the raw event.

The second strand is the *strategy and management* literature on the consequences and interpretation of downsizing. Cascio (1993) and Love and Nohria (2005) distinguish reactive, cost-cutting “retrenchment” from proactive, strategic “reorientation,” and argue that the two have different performance consequences; Guthrie and Datta (2008) show the effect of downsizing on firm performance is moderated by industry conditions. This literature is precisely about *motive*. Our contribution is to test whether motive is legible in the one motive-like field available in administrative data—the plant-closure-vs-layoff category—and to show, with a power calculation, that it is not: the machine flag is both a point-null and too imprecise to rule out the motive effect this literature predicts. Measuring motive therefore requires the narrative, which motivates the hand-coding step.

The third strand is *event-study methodology and the interpretation of null results*. We follow the standard market-model event-study framework (MacKinlay, 1997; Brown and Warner, 1985) and, because event-induced variance and cross-sectional correlation can distort conventional *t*-tests (Boehmer et al., 1991; Corrado, 1989; Kolari and Pynnönen, 2010), we complement parametric inference with randomization inference (Young, 2019), which is valid under the sharp null without asymptotic assumptions on the number of clusters. We also follow the guidance that a statistically insignificant estimate should be interpreted through its power and minimum detectable effect rather than treated as proof of no effect (Abadie, 2020): Section 10 states the motive null as a bound (5.1pp at 80% power) rather than a bare “insignificant,” which is what makes the case for hand-coding quantitative rather than rhetorical.

3 Hypotheses

Layoff announcements carry two offsetting signals (Palmon et al., 1997; Chen et al., 2001). A cut can be *good news*—evidence that management is disciplined and is lowering the cost base (an “efficiency” signal)—or *bad news*—evidence that demand for the firm’s product is falling (a “demand” signal). Which dominates should depend on how material the cut is and on why it is being made.

H1 (Average reaction). The average CAR around a WARN mass-layoff notice is non-zero. (Sign is an empirical question: the efficiency and demand signals push in opposite directions.)

A cut that is large relative to the firm is more likely to be read as a demand signal about the whole franchise, whereas a cut that is trivial relative to the firm is noise.

H2 (Materiality). CARs are decreasing in the size of the layoff *relative to the firm* (relative-layoff ratio), and, equivalently, increasing in firm size. Absolute headcount, which does not scale by the firm, should matter less.

Finally, the motive. A layoff that *reallocates* resources away from a declining business (often a plant closure) toward better uses may be read differently from a defensive, across-the-board *cost-cutting* layoff (Love and Nohria, 2005; Nixon et al., 2004).

H3 (Motive). CARs differ by layoff motive. If the administrative closure-vs-layoff flag captures motive, closures and ongoing-operation layoffs should earn different CARs; if it does not, the flag will be uninformative and motive must be measured from the announcement itself.

4 Data and Variables

WARN notices. The WARN Act requires employers with 100+ workers to give 60 days’ notice of a plant closing or mass layoff; states publish the notices. We use three machine-

readable state sources: the California EDD detailed WARN reports (four fiscal-year files, 07/2022–06/2026, parsed from the official PDF/XLSX), the Texas Workforce Commission WARN dataset (`data.texas.gov`), and the Oregon WARN dataset (`data.oregon.gov`). Each notice carries the filer name, the notice date, the number of employees affected, the location, and (in CA/OR) a layoff/closure type. Pooling and restricting to 2022–2025 gives 6,560 notices from 3,450 distinct filer names (CA 5,548; TX 815; OR 190; a few notices in other states appear in the Oregon cross-file).

Matching to listed firms. WARN filers are operating names at a job site (“Amgen,” “Applied Materials, Inc. (557 E. Ca...)”). We build a name universe of U.S. common stocks active 2021–2026 from CRSP (`stocknames`) plus Compustat legal names (`company`) linked to permno via the CRSP/Compustat link, normalize both sides (dropping legal suffixes, “dba” trade names, and store/site numbers), and fuzzy-match each filer to the nearest listed name with a token-set score, keeping a match only when it clears a strict score *and* a leading-anchor test (the filer name and the listed name share their first distinctive token). This high-precision pass matches 738 distinct filer names to 412 listed firms. It deliberately *misses* brand/subsidiary filings (“Pixar”→Disney, “Optum”→UnitedHealth); resolving those and verifying the automated matches is a hand task (`STUDENT_TASKS.md`). Because the event study rests entirely on these matches, this precision-over-recall choice is deliberate: a false match injects an unrelated firm’s return into the sample, whereas a missed match merely omits an event. The hand-verification and brand-rescue steps reserved for the student directly raise recall without sacrificing precision.

From notices to events. Firms often file several site notices within days. We cluster a firm’s notices into *events*: a new event begins when a notice is more than 20 calendar days after the current event’s first notice; the event date is the first notice in the cluster and headcount is the cluster sum. This yields 852 firm-events at 412 firms. Merging Compustat fundamentals (most recent fiscal year before the event: assets, employees, sales, market cap,

industry) and CRSP returns leaves 615 firm-events at 328 firms with a valid market model.

Variables. *Relative layoff* = cluster headcount / total firm employees. *Firm size* = $\ln(\text{assets})$ (and $\ln(\text{market cap})$ in robustness). *Closure* is an indicator that the filing’s type field contains “closure” (our crude reallocation/exit proxy). *Tech* is an indicator for computer/electronics/software SIC codes. In the extended heterogeneity analysis we add three further *observable* moderators, all measured at the pre-event fiscal year and all deliberately orthogonal to the reserved hand-coded motive: market-to-assets (market cap/assets, a glamour/growth proxy), asset turnover (sales/assets), and a goods-producing (non-tech manufacturing) indicator. Continuous predictors are winsorized at 1/99 and standardized, so a coefficient is the effect of a one-SD increase, in CAR points. Table 1 reports summary statistics: the median WARN event lays off 86 workers, 0.8% of the firm’s workforce, at a firm with \$8B in assets—confirming these are locally large but firm-level small cuts at sizable public companies.

5 Research Design and Inference

Event study. Day 0 is the first trading day on/after the notice date. We estimate the market model $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$ over $[-252, -46]$ against the CRSP value-weighted index and cumulate abnormal returns over several windows. The cross-sectional regressions take a CAR as the dependent variable and firm/layoff characteristics as regressors, with standard errors clustered by firm (some firms file multiple events).

Inference. Cross-sectional event-study t -tests can be distorted by event-induced variance and by cross-sectional correlation among abnormal returns (Boehmer et al., 1991; Corrado, 1989; Kolar and Pynnönen, 2010). We take three precautions beyond clustered standard errors. First, we report a pre-event $[-10, -2]$ placebo window, which should be zero if the reaction is news rather than anticipation or a pre-trend. Second, we subject the two

coefficients that carry the paper—the materiality slope and the machine-motive (closure) coefficient—to *randomization inference*: we permute the regressor across events 2,000 times and compare the actual coefficient to the permutation distribution, yielding a design-based p -value that is valid under the sharp null without asymptotic assumptions (Young, 2019). Third, we re-derive the CARs under three alternative abnormal-return benchmarks straight from the raw daily returns, so the result cannot be an artifact of the market-model specification. Finally, we convert “insignificant” into a bound by reporting the minimum detectable effect at 80% power for each key coefficient (Abadie, 2020). None of these steps uses the reserved hand-coded motive; all use observable data only.

6 Main Results

The average reaction (H1). Table 2 and Figure 1 give the event-study result. The market marks the stock *down* after a WARN notice: mean CAR of -0.9% over $[0, +1]$, -1.3% over $[0, +2]$ ($t = -2.22$), and -1.6% over $[0, +5]$ ($t = -2.41$). The drift is post-event—the $[-10, -2]$ placebo is an insignificant -0.18% —consistent with the notice conveying news rather than being anticipated. But the reaction is not universal: fewer than half of events (49%) have negative CAR $[0, +5]$, so the negative mean reflects a tail of badly-received layoffs rather than a uniform markdown. H1 is supported (the average is significantly negative) with the caveat that the median firm’s stock barely moves.

Materiality, not headcount (H2). Table 4 is the core cross-section. Relative layoff size is the variable that matters: alone (column 1) a one-SD larger relative layoff is worth -3.3pp of CAR ($t = -2.77$). Firm size enters with the mirror-image sign: a one-SD larger firm is worth $+3.1\text{pp}$ ($t = 3.34$, column 3). Absolute headcount is a precise zero (column 2, $R^2 = 0.000$, $t = 0.46$): the market does not react to the *number* laid off, only to its *proportion* of the firm. Because relative-layoff and firm size are two encodings of the same materiality, they partly compete when entered together (column 4: relative layoff -2.6pp ,

$t = -1.87$; size +1.5pp, $t = 1.53$), but the materiality channel is unmistakable. The subgroup means in Table 3 and Figure 2 tell the same story without any modeling: small firms -3.9% ($t = -2.86$) versus large firms $+0.4\%$ (ns); top-half relative layoffs -3.5% ($t = -2.70$) versus bottom-half $+0.2\%$ (ns). H2 is strongly supported.

Motive is not in the flag (H3). The closure indicator—our administrative proxy for strategic exit/reallocation as opposed to cost-cutting—is economically and statistically zero throughout. Closures earn -1.5% and ongoing-operation layoffs -1.6% (Table 3); in the regression the closure dummy is -0.001 ($t = -0.07$, Table 4 column 5), and it stays insignificant in every robustness cut (Table 10). The market clearly *does* discriminate among layoffs—by materiality—but the closure-vs-layoff category available in administrative data does not capture the discrimination. H3 is *not* supported *with the machine proxy*. As Section 10 shows, this is not because we have precisely estimated a zero motive effect; it is because the machine proxy is too coarse to detect one. That is the paper’s pivotal negative result: if motive matters (as theory and the strategy literature argue), it is encoded in the announcement narrative, not in the WARN type field, and only hand-coding can recover it.

7 Heterogeneity: Where the Bite Is

The punishment is a small-firm phenomenon. Table 5 shows the materiality effect is concentrated among small firms. Interacting relative layoff with a below-median-size dummy, the relative-layoff slope is a positive $+0.041$ ($t = 2.08$) among large firms and the small-firm interaction is -0.065 ($t = -2.86$)—so among large firms a bigger relative cut is, if anything, mildly *rewarded* (an efficiency read), while among small firms it is punished (a demand read). Neither the closure nor the tech interaction is significant. That a given relative cut flips sign with firm size is itself suggestive that *motive* is being inferred differently across firms—and is a further reason to measure motive directly.

Not concentrated in any other observable subgroup. If the materiality effect were really a proxy for some other firm characteristic, it should concentrate in the corresponding observable subgroup. Table 6 interacts the standardized relative-layoff ratio with four additional pre-event, observable moderators—high market-to-assets (glamour/growth), high asset turnover, an early-sample (2022) indicator, and a goods-producing (non-tech manufacturing) indicator—always controlling for firm size. In every case the materiality slope is negative (ranging -0.018 to -0.026) and the interaction is economically small and statistically indistinguishable from zero (the largest $|t|$ on any interaction is 0.58). The materiality effect is thus a broad phenomenon, not a growth-firm, capital-intensity, sample-period, or industry-specific artifact. Crucially, we do *not* interact the reaction with the hand-coded motive here—that dimension is unobservable to us and is the student’s reserved contribution; the moderators above are observable firm characteristics chosen precisely because they are orthogonal to that reserved test.

8 Randomization Inference

Cross-sectional CAR regressions with clustered standard errors rely on asymptotics in the number of clusters, and event-induced variance can inflate conventional test statistics (Boehmer et al., 1991; Kolar and Pynnönen, 2010). Table 7 and Figure 3 report design-based inference from 2,000 permutations of the regressor under the sharp null of no cross-sectional relation. The materiality slope (-0.024 in the size-controlled specification) lies far outside the permutation distribution (95% permutation interval $[-0.014, +0.013]$; design-based $p = 0.001$): the relation between relative layoff size and CARs is not a clustered-SE artifact. By contrast, the machine-motive (closure) coefficient (-0.001) sits essentially at the center of its permutation distribution ($p = 0.92$, 95% interval $[-0.031, +0.030]$)—the motive null is robust to design-based inference. One caveat sharpens rather than weakens the reading: our permutation reassigns the regressor at the event level and so ignores the within-firm dependence

that the clustered standard errors (328 firms, 615 events) are designed to handle. Randomization inference and the clustered t therefore *bracket* the true inference—the design-based $p = 0.001$ is a lower bound on significance and the size-controlled clustered $t = -1.81$ an upper bound—and the materiality effect is significant on the unconditional clustered $t = -2.77$ as well. Randomization inference thus corroborates both halves of the paper: the materiality effect is a genuine signal, and the machine-motive null is real rather than a failure of the parametric standard errors. It does not, however, tell us whether the motive null is a *tight* bound; that is the question the power analysis in Section 10 answers, and its answer is what makes the hand-coding necessary.

9 Alternative Abnormal-Return Benchmarks

The headline CARs use a firm-specific market model. To rule out that either the average reaction or the materiality slope is an artifact of that benchmark, we re-derive CAR[0, +5] straight from the raw CRSP daily returns under three alternatives: market-adjusted returns (abnormal return = $R_{it} - R_{mt}$, i.e. β forced to one), S&P-500-adjusted returns ($R_{it} - R_t^{\text{S\&P}}$), and mean-adjusted returns (R_{it} less the firm’s estimation-window mean). As a correctness check, the code first rebuilds the market-model CAR from the stored α, β and confirms it matches the base CARs to machine precision ($\max |\text{diff}| < 10^{-15}$ across 571 events). Table 8 reports, for each benchmark, the average CAR (H1) and the materiality slope (H2). The average negative reaction is not only robust but *stronger* against the two index benchmarks that do not net out firm β : -2.0% ($t = -3.03$) market-adjusted and -2.0% ($t = -3.06$) S&P-adjusted, versus -1.6% ($t = -2.41$) for the market model. It is weaker and marginally insignificant under the mean-adjusted model (-1.0% , $t = -1.53$)—expected, because mean-adjustment ignores the generally negative market drift over some event windows and so is the noisiest benchmark. The materiality slope is stable and negative across all four benchmarks (-0.020 to -0.027 ; $|t|$ up to 2.08). The core results survive the choice of abnormal-return

model.

10 Is the Motive Null Informative? Power and Minimum Detectable Effects

A null is only interesting if the design could have detected a real effect, and here the answer differs sharply between the materiality slope and the motive proxy. Table 9 reports, for each key coefficient, the minimum detectable effect at 80% power, $MDE = (z_{0.975} + z_{0.80}) \widehat{SE} \approx 2.80 \widehat{SE}$. For the materiality slope the estimate (-2.5 pp per SD in the size-controlled specification, -3.3 pp unconditionally) is comparable to its ≈ 3.9 pp 80%-power MDE: the design is powered to detect roughly the effect it actually finds. That the size-controlled clustered t is only -1.81 reflects the mechanical relative-layoff/size collinearity flagged above, not a fragile effect—the unconditional slope is significant at $t = -2.77$ and the randomization-inference p is 0.001 (Section 8). The average reaction is likewise a clear but not high-powered signal (estimate -1.6 pp, MDE 1.9pp; significant at $t = -2.41$ because significance requires $1.96 \times SE$ while the 80%-power MDE is $2.80 \times SE$). The machine-motive proxy is a different case entirely. Its coefficient is a near-perfect zero (-0.1 pp), but its standard error is large (1.8pp), so the smallest motive effect it could have detected at 80% power is *5.1 percentage points*—larger than the entire average market reaction to a layoff and far above any plausible motive-driven CAR difference in the strategy literature. In other words, the closure flag does not deliver an informative null: it is both a point estimate of zero *and* too imprecise to rule out an economically meaningful motive effect. This is the quantitative case for hand-coding. A sharper, narrative-based motive measure would (i) replace a coarse binary that mixes cost-cutting and reallocation within the “closure” category with a clean strategic-vs-defensive split, and (ii) plausibly reduce the noise that inflates the current MDE, so that a true motive effect of, say, 1–3pp—below the current 5.1pp detection threshold but consistent with the layoff literature—could actually be detected. The design has been built to receive

exactly that measure.

11 Robustness

Table 10 re-estimates the full specification under alternative choices. The relative-layoff effect is stable and negative across dependent variables (CAR[0, +2]: -0.031 , $t = -2.01$; CAR[−1, +5]: -0.028 , $t = -2.04$), when 2022 events are dropped (-0.030 , $t = -1.73$), when we restrict to U.S.-incorporated firms (-0.025 , $t = -1.75$), and when firm size is measured by market capitalization instead of assets (-0.029 , $t = -2.15$). The firm-size coefficient is consistently positive, attenuating to insignificance only when relative layoff (its near-mirror) is also in the model—an expected multicollinearity, not a fragility. The closure (motive) proxy is insignificant in all six columns. Together with the alternative-benchmark table (Table 8) and the randomization-inference table (Table 7), the materiality result is robust to window, sample, size proxy, abnormal-return benchmark, and inference method; the null on the machine-coded motive is equally robust—and, per Section 10, equally uninformative as a bound.

12 Conclusion

Using a new matched WARN-to-CRSP sample, we find that the market’s reaction to a mass-layoff notice is, on average, a modest negative -1.6% over five days (robust, and stronger, under alternative benchmarks), but that this average is almost entirely about *materiality*: layoffs that are large relative to a small firm are punished (-3.9%), while the same administrative event at a large firm is a non-event. The number of workers laid off, per se, carries no information, and randomization inference confirms the materiality effect is a genuine signal ($p = 0.001$) rather than a standard-error artifact. Most importantly for the question in our title, the one motive proxy available in administrative data—plant closure versus ongoing-operation layoff—does *not* distinguish the CARs, and the power analysis

shows this is not a tight null: the machine proxy could only have detected a motive effect of 5.1 percentage points, so it cannot rule out the economically meaningful motive effects the strategy literature predicts. Whether a layoff is cost-cutting or strategic reallocation is not legible in the WARN category field. The natural and necessary next step, which this paper is designed to enable, is to read the announcements and *hand-code* the motive, then re-run the cross-section on it: only then can we answer whether the market truly punishes cost-cutting and rewards reallocation, or vice versa. Limitations include the three-state footprint (CA, TX, OR), the lead-name matching (hand-verification pending), the 2022–2024 return window, and the coarse motive proxy—each of which the student work is built to address.

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Table 1: Summary statistics

Variable	N	Mean	SD	P25	Median	P75
CAR[0,+5]	571	-0.0159	0.1579	-0.0349	-0.0010	0.0284
CAR[0,+2]	571	-0.0131	0.1412	-0.0242	0.0013	0.0215
CAR[-1,+1]	573	-0.0044	0.1602	-0.0238	-0.0007	0.0226
Layoff headcount	615	177.6650	332.2369	52.0000	86.0000	175.5000
Rel. layoff (head/emp)	605	0.1481	1.6386	0.0014	0.0079	0.0431
ln(Assets)	614	8.9155	2.2043	7.1792	8.9557	10.6545
ln(Mkt cap)	614	8.9056	2.5687	7.1311	8.9423	10.8217
Closure (0/1)	615	0.2260	0.4186	0.0000	0.0000	0.0000
Tech (0/1)	615	0.3642	0.4816	0.0000	0.0000	1.0000

Notes: 615 matched firm-events, 2022–2024. CARs are market-model (CRSP value-weighted index, estimation window $[-252, -46]$). Relative layoff = event headcount / firm employees; firm fundamentals are the most recent Compustat fiscal year before the event.

Table 2: Cumulative abnormal returns around WARN notices, by window

Event window	Mean CAR (%)	Median (%)	<i>t</i> -stat	% > 0	N
[-1,+1]	-0.44	-0.07	-0.65	50%	573
[0,+1]	-0.88	0.10	-1.69*	51%	571
[0,+2]	-1.31	0.13	-2.22**	52%	571
[0,+5]	-1.59	-0.10	-2.41**	49%	571
[-1,+5]	-1.14	-0.08	-1.45	50%	573
[-10,-2] placebo	-0.18	0.07	-0.32	51%	579

Notes: Day 0 = first trading day on/after the WARN notice date. $[-10, -2]$ is a pre-event placebo. *t*-statistics test the mean CAR = 0. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Mean CAR[0,+5] by subgroup

Subgroup	Mean CAR[0,+5] (%)	Median (%)	<i>t</i> -stat	N
Closure	-1.51	0.20	-0.91	136
Ongoing-op layoff	-1.62	-0.23	-2.32**	435
Tech	-0.38	-0.01	-0.42	206
Non-tech	-2.28	-0.23	-2.54**	365
Small firm (below-median assets)	-3.85	-1.16	-2.86***	268
Large firm (above-median assets)	0.37	0.28	1.13	302
Large rel. layoff (top half)	-3.45	-0.84	-2.70***	271
Small rel. layoff (bottom half)	0.17	0.23	0.37	296

Notes: Small/large firm split at median $\ln(\text{assets})$; relative-layoff split at the median. Closure vs. ongoing-operation layoff from the WARN type field (our crude motive proxy).

Table 4: Cross-section of WARN-notice CARs: materiality, not headcount or motive

	(1) Rel. layoff	(2) $\ln(\text{Headcount})$	(3) $\ln(\text{Assets})$	(4) All three	(5) +Motive/Tech/Yr FE
Rel. layoff (head/emp), z	-0.0335*** (-2.77)			-0.0256* (-1.87)	-0.0253* (-1.82)
$\ln(\text{Layoff headcount})$, z		0.0030 (0.46)		0.0057 (0.89)	0.0052 (0.74)
$\ln(\text{Assets})$, z			0.0308*** (3.34)	0.0153 (1.53)	0.0154 (1.57)
Closure (0/1)					-0.0013 (-0.07)
Tech (0/1)					0.0058 (0.47)
Year FE	No	No	No	No	Yes
Observations	567	568	570	564	564
R^2	0.043	0.000	0.037	0.052	0.053

Notes: DV is CAR[0, +5]. Continuous predictors winsorized 1/99 and standardized (coefficient = effect of a one-SD increase, in CAR points). *Closure* is the plant-closure indicator (proxy for strategic exit/reallocation vs. cost-cutting). *t*-statistics clustered by firm in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Heterogeneity: the punishment is a small-firm phenomenon

Specification	Rel. layoff (z)	Interaction	$\ln(\text{Assets})$ (z)	N	R^2
Closure x Rel.layoff	-0.0215 (-1.55)	-0.0103 (-0.33)	0.0168* (1.76)	567	0.052
Size x Rel.layoff	0.0405** (2.08)	-0.0647*** (-2.86)	0.0255* (1.81)	567	0.054
Tech x Rel.layoff	-0.0275* (-1.79)	0.0213 (1.23)	0.0168* (1.77)	567	0.054

Notes: DV is CAR[0, +5]. Each row adds an interaction of the standardized relative-layoff ratio with, respectively, the closure dummy, a below-median-size dummy, and the tech dummy. Firm-clustered *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Extended heterogeneity: the materiality effect is not concentrated in any observable subgroup

Moderator M (observable, pre-event)	Rel. layoff (z)	Rel. $\times M$	M	N
High market-to-assets (glamour)	-0.0255 (-1.55)	+0.0042 (+0.18)	-0.0069 (-0.52)	567
High asset turnover	-0.0184 (-1.17)	-0.0168 (-0.58)	+0.0101 (+0.56)	567
Early sample (2022 event)	-0.0260 (-1.54)	+0.0058 (+0.25)	+0.0020 (+0.10)	567
Goods-producing (non-tech mfg.)	-0.0237* (-1.69)	-0.0002 (-0.01)	-0.0044 (-0.31)	567

Notes: DV is CAR[0, +5]. Each row is a separate regression that interacts the standardized relative-layoff ratio with one pre-event, observable moderator M , always controlling for $\ln(\text{assets})$. Moderators: high market-to-assets (glamour/growth), high asset turnover, an early-sample (2022 event) indicator, and a goods-producing (non-tech manufacturing) indicator. Reported are the relative-layoff slope, the relative $\times M$ interaction, and M itself, with firm-clustered t -statistics in parentheses. The reserved hand-coded motive is deliberately *not* used as a moderator. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Randomization inference (2,000 permutations of the regressor)

Coefficient (DV = CAR[0,+5])	Actual	Perm. 95% interval	RI p -value
Rel. layoff (materiality), z	-0.0242	[-0.0135, +0.0128]	0.0010
Closure (machine motive proxy)	-0.0015	[-0.0311, +0.0296]	0.9230

Notes: DV is CAR[0, +5]. The regressor is permuted across events 2,000 times under the sharp null of no cross-sectional relation; the same OLS specification is re-estimated each time. The materiality slope is from the size-controlled specification (relative layoff + $\ln(\text{assets})$); the closure coefficient is from the full specification. The “Perm. 95% interval” is the 2.5–97.5 percentile of the permutation distribution; the RI p -value is the share of permutations with $|\text{coefficient}| \geq |\text{actual}|$. See Figure 3.

Table 8: Alternative abnormal-return benchmarks

Abnormal-return benchmark	Mean CAR (%)	t	Rel. layoff (z)	$\ln(\text{Assets})$ (z)	N
Market model (base)	-1.59	-2.41**	-0.0242* (-1.81)	+0.0165* (+1.73)	567
Market-adjusted ($\beta=1$)	-2.01	-3.03***	-0.0274** (-2.08)	+0.0193** (+1.98)	567
S&P 500-adjusted	-2.03	-3.06***	-0.0271** (-2.05)	+0.0196** (+2.01)	567
Mean-adjusted	-1.02	-1.53	-0.0197 (-1.48)	+0.0176* (+1.83)	567

Notes: CAR[0, +5] re-derived from raw CRSP daily returns under four benchmarks: the firm-specific market model (base); market-adjusted ($R_{it} - R_{mt}$, $\beta=1$); S&P-500-adjusted ($R_{it} - R_t^{\text{S\&P}}$); and mean-adjusted (R_{it} less the firm’s estimation-window mean). “Mean CAR” and its t test H1; the relative-layoff and $\ln(\text{assets})$ slopes (size-controlled specification, firm-clustered t in parentheses) test H2. The rebuilt market-model CAR reconciles with the base CARs to machine precision. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Minimum detectable effects at 80% power

Coefficient	Estimate (pp)	SE (pp)	MDE 80% (pp)
Rel. layoff (materiality), z	-2.53	1.39	3.89
$\ln(\text{Assets})$, z	+1.54	0.98	2.75
Closure (machine motive proxy)	-0.13	1.82	5.09
Average CAR[0,+5]	-1.59	0.66	1.85

Notes: $\text{MDE}(80\%) = (z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.80 \times \text{SE}$, in CAR percentage points. Coefficients are from the full cross-sectional specification (Table 4, column 5); the average CAR row uses the one-sample standard error. A *small* MDE relative to the estimate (materiality; average CAR) indicates a well-identified effect; a *large* MDE relative to plausible effect sizes (the closure/motive proxy, 5.1pp) indicates an uninformative null—the design could not have detected an economically meaningful motive effect with the machine proxy, motivating the hand-coded motive measure.

Table 10: Robustness

Specification	Firm size (z)	Rel. layoff (z)	Closure	N	R^2
DV=CAR[0,+5] (base)	0.0152 (1.52)	-0.0252* (-1.84)	-0.0015 (-0.08)	564	0.053
DV=CAR[0,+2]	0.0032 (0.38)	-0.0308** (-2.01)	-0.0120 (-0.64)	564	0.054
DV=CAR[-1,+5]	0.0102 (0.92)	-0.0280** (-2.04)	0.0067 (0.25)	566	0.036
Exclude 2022	0.0104 (1.23)	-0.0302* (-1.73)	-0.0126 (-0.61)	469	0.061
US firms only	0.0169 (1.50)	-0.0249* (-1.75)	-0.0032 (-0.16)	512	0.055
$\ln(\text{Mktcap})$ for size	0.0088 (0.83)	-0.0290** (-2.15)	-0.0001 (-0.00)	564	0.049

Notes: Full specification (relative layoff, $\ln$ headcount, firm size, closure, tech) re-estimated under alternative dependent variables, samples, and size proxies. Only the size, relative-layoff, and closure coefficients are shown. Firm-clustered t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

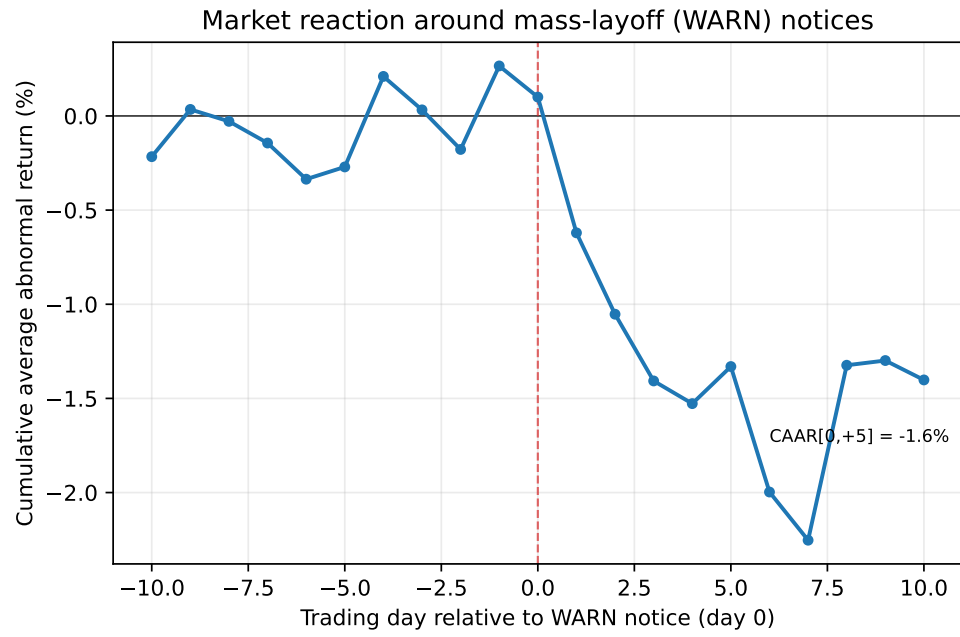


Figure 1: Cumulative average abnormal return in event time around WARN mass-layoff notices (615 firm-events, 2022–2024). The reaction is post-event; the pre-event path is flat.

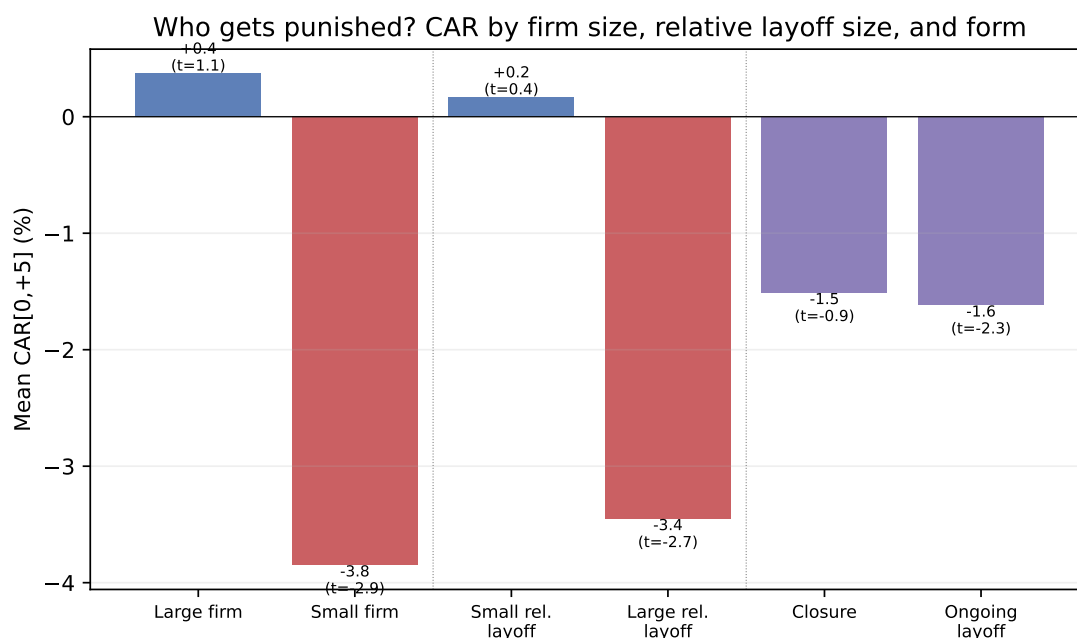


Figure 2: Mean CAR[0, +5] by firm size, relative layoff size, and layoff form. The market punishes layoffs that are large relative to a small firm; the closure (vs. ongoing layoff) form—our motive proxy—does not distinguish the reaction.

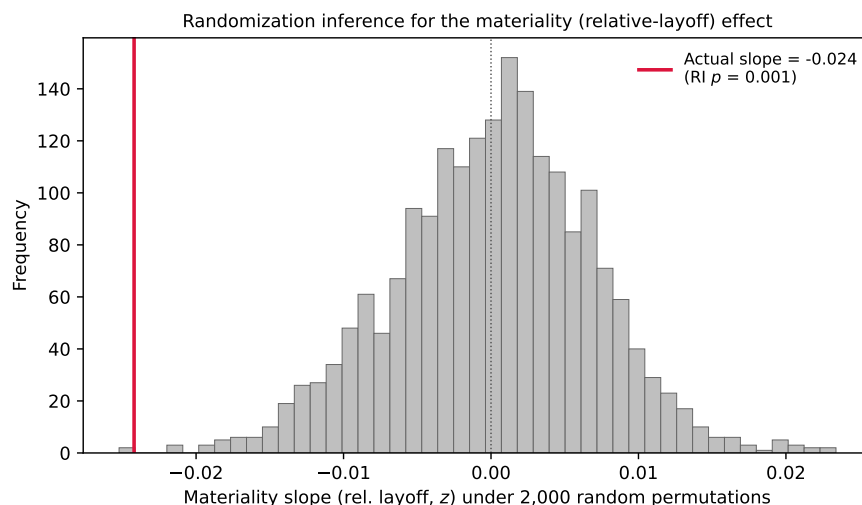


Figure 3: Randomization inference for the materiality (relative-layoff) slope. Histogram: the distribution of the slope under 2,000 random permutations of the relative-layoff regressor. The actual estimate (red line) lies far in the left tail (RI $p = 0.001$), so the materiality effect is a genuine cross-sectional signal rather than a clustered-standard-error artifact.